TECHNICAL LEARNING GUIDE · OCTOBER 2026

Code Changes and Learning Guide

A practical walkthrough of the PlanZ deployment and fresh-browser fixes: what failed, why it failed, how the code now handles it, and how to debug similar problems yourself.

DEPLOYMENT BUG

Task module path escaped the project

The browser requested a shared module from the domain root instead of the repository's GitHub Pages subpath. That stopped the task page module from loading.

FRESH-SESSION BUG

Optional user data was assumed to exist

Voice, name, and avatar handlers dereferenced a missing localStorage record and threw on a fresh origin.

ASSET BUG

Profile image extension was wrong

The profile stylesheet requested a PNG while the repository contains a JPG.

PURPOSE

Make behavior match real browser state

Keep the task module inside the published site, tolerate empty browser storage, and load the existing profile asset.

Files changed

File	Change
`javascript2.js`	Corrected one import URL and guarded user-data writes in voice, name, and avatar handlers.
`Profile-style.css`	Changed the sticker image reference from `.png` to the tracked `.jpg`.

> **Production status:** the live browser observation that confirmed the import 404 was made before the fix was deployed. The changes are in commit `48a6ae3`; this guide does not claim the current Pages deployment was rechecked against that commit.

Project: **PlanZ**
Repository: `zakariawebdesigner-png/Z-Plan`
Pages URL: `https://zakariawebdesigner-png.github.io/Z-Plan/`
Commit examined: `48a6ae3b121ab434f5f5357badd53b807da6cacb`

1. The task module left the Pages site

CONFIRMED BUG

Problem. The deployed task page could not load its main JavaScript module. Because the browser failed to resolve one imported module, the task entry module did not execute and its event listeners were never registered.

Cause. The importing file, `javascript2.js`, sits at the project root. Its old path began with two parent-directory segments:

```
import { StyleMode } from '../../Shared/Ui/Theme-Modes.js';
```

On GitHub Pages, the project is published under `/Z-Plan/`. Resolving that path from the root script escapes the project directory and asks the server for `/Shared/Ui/Theme-Modes.js`.



Evidence. In headless Chrome on the live site, Network recorded a 404 for `https://zakariawebdesigner-png.github.io/Shared/Ui/Theme-Modes.js`. The task module failed to load. After correcting the import, the local browser run created a task successfully.

Solution. Resolve the import from the file's actual directory:

```
import { StyleMode } from './Shared/Ui/Theme-Modes.js';
```

Why it works. A relative module specifier is resolved against the importing module's URL, not against the HTML document. From `/Z-Plan/javascript2.js`, `./Shared/…` stays under `/Z-Plan/`.

> LEARNING POINT
>
> When a module request fails in production, inspect the final URL in DevTools Network. Count path segments from the importing file; don't guess from the page URL.

Relevant final code: `javascript2.js`, import near line 580. Deployment context: the repository Pages workflow publishes the repository root.

2. A new site origin has empty storage

CONFIRMED EDGE-CASE BUG

Problem. Clicking a voice option with no saved user record threw a `TypeError`. The same assumption existed in the quick-profile name and avatar handlers.

Cause. `localStorage.getItem("UserInfo_Object")` returns `null` when the key is absent. The old code then immediately assigned a property to that value.

```
let user = JSON.parse(localStorage.getItem("UserInfo_Object"));
user.AiVoice = card.id; // throws when user is null
```

Why this shows up after deployment. Browser storage is separated by *origin*: scheme + host + port. Data saved on `http://localhost:…` is not available at `https://zakariawebdesigner-png.github.io`. A first visit to the Pages site therefore starts with a fresh storage area.

Solution. Only update account data when it exists. The voice itself still plays; the code simply skips saving a preference when there is no account record.

```
let user = JSON.parse(localStorage.getItem("UserInfo_Object"));
if (user) {
    user.AiVoice = card.id;
    localStorage.setItem("UserInfo_Object", JSON.stringify(user));
}
AiVoice(voice.audio);
```

The same early-return guard now protects name and avatar updates. This avoids both the exception and accidentally creating a partial user record.

> EXPECTED LIMITATION
>
> These changes do not sync localhost data to GitHub Pages. Each origin keeps its own data. Sign up on the deployed site to create data there; cross-device accounts require a backend or an explicit export/import feature.

> LEARNING POINT
>
> Treat storage reads as optional input. Check for missing data before dereferencing it, and decide whether the UI can continue without persisting that preference.

3. The profile sticker used the wrong extension

CONFIRMED ASSET BUG

Problem. The profile CSS requested `photos/AddedNotes-Sticker.png`, but that file is not in the repository. The tracked image is `photos/AddedNotes-Sticker.jpg`.

Solution. In `Profile-style.css`, change the URL extension to `.jpg`. CSS asset URLs resolve relative to the stylesheet, so this becomes `/Z-Plan/photos/AddedNotes-Sticker.jpg` on the deployed project site.

Verification. Before the change, the browser reported a 404 for the PNG path. After the change, the local profile page produced no failed asset requests.

File-by-file guide

File	Role	What changed and why
`javascript2.js`	Entry script for task interactions and quick profile settings.	Fixed one shared-module URL; added missing-data checks to four voice choices and the name/avatar handlers.
`Profile-style.css`	Styles the profile page, including sticker artwork.	Points the sticker rule to the exact JPG filename tracked in `photos/`.

No HTML structure, visual design, dependency, or build configuration was changed. The guards preserve existing behavior when a saved user record is present.

How the task module loads



If any static import in this graph fails, the browser rejects evaluation of the importing module. The page may still render its HTML and CSS, which can make the failure look like “buttons do nothing.”

Relative module URLs

Before:

```
import { StyleMode } from '../../Shared/Ui/Theme-Modes.js';
```

After:

```
import { StyleMode } from './Shared/Ui/Theme-Modes.js';
```

1. The browser fetches `javascript2.js` from the project root.
2. It resolves the import relative to that script's own URL.
3. `../../` climbs out of the repository's Pages subpath.
4. `./` starts from the same project root and reaches the checked-in module.

> **Debugging clue:** a 404 on an imported module is a graph failure. Check the exact request URL, then check the import path relative to the module named in the parent request.

Null guards and graceful behavior

```
// Read the account record; it may not exist in a fresh browser.
const user = JSON.parse(localStorage.getItem("UserInfo_Object"));
if (user) {
    user.AiVoice = card.id;
    localStorage.setItem("UserInfo_Object", JSON.stringify(user));
}
AiVoice(voice.audio);
```

Step by step:

1. `getItem` returns either a JSON string or `null`.
2. `JSON.parse` turns the string into an object; parsing `null` yields `null`.
3. The `if` block runs only when there is an object to update.
4. Audio playback remains outside the guard, so choosing a voice still has an immediate effect without an account.

The name and avatar handlers use an early return for the same reason: stop only the account-specific part when its required data is absent.

06 · CONCEPTS TO CARRY FORWARD

Three ideas behind these fixes

1. A relative URL has a base

For an ES module import, the base is the importing JavaScript file. For a CSS `url(...)`, the base is the stylesheet. For an HTML link, the base is the document URL.

Always identify the base file before counting `../` segments. A repository site such as `owner.github.io/repo/` has an extra path segment that a user site at `owner.github.io/` does not.

2. Browser storage belongs to an origin

`http://localhost:5500`, `http://localhost:3000`, and `https://owner.github.io` are different origins. Their localStorage values are isolated by the browser. That is a browser security boundary, not a deploy-copy mechanism.

Use a fresh browser profile during testing to reveal assumptions hidden by old local data. Do not “fix” this by clearing storage: that deletes data and does not create cross-origin sync.

3. Modules load as a dependency graph

A page’s entry module imports other files, which may import more files. If a required import returns 404 or has an invalid export, the importing module does not run. Handlers declared in that module are therefore absent, even while the HTML remains visible.

Network errors often explain silent-looking UI failures. Check both Console and Network; then follow the failed request back to its import statement.

> **PROFESSIONAL HABIT**
>
> Test the state your code must support, not only the state left behind by your own browser. “No saved user yet” is a normal state for a first-time visitor.

A simple path check

```
Importer: /Z-Plan/javascript2.js
Specifier: ./Shared/Ui/Theme-Modes.js
Request:   /Z-Plan/Shared/Ui/Theme-Modes.js  ✓
```

A CSS path is resolved differently: check it from the CSS file’s URL, not from `Profile.html`.

Prevent similar failures

Avoid	Why it hurts	Check instead
Guessing an import path from the HTML page.	Imports resolve from the importing module's URL.	Inspect the failed Network URL and calculate from the parent script.
Assuming localStorage always has a record.	New origins, private sessions, and cleared data start empty.	Test with no saved key; guard before reading object properties.
Changing a filename extension without checking the repository.	Linux-hosted paths must match real filenames exactly.	Compare path, capitalization, and extension with the tracked asset.
Assuming a rendered page means its JavaScript ran.	HTML/CSS can display while a module graph has failed.	Check Console, Network, and one key interaction per page.

Verification actually performed

Check	Result
Live Pages before correction: landing, sign-up, task page.	Landing and sign-up had no captured JavaScript errors. Task page requested the shared module outside `/Z-Plan/` and received 404.
Local corrected browser: startup, task creation, note creation, routine creation.	Each created one item; no application JavaScript exceptions or failed local requests.
Fresh-storage voice/name/avatar controls.	No exceptions after guards; no partial user record was created.
Sign-up → loader → task page.	Reached the task page with temporary sample user data and no JavaScript exceptions.
Profile page after image correction.	No failed profile asset requests or JavaScript exceptions.
Syntax / whitespace checks.	`node --check javascript2.js` and `git diff --check` passed.

Headless synthetic clicks cannot grant trusted user activation for audio. Routine tests logged browser autoplay rejections from the test harness; this does not prove a real click fails.

> **Still to verify:** inspect the live Pages site after the latest commit is deployed. The live task failure was recorded before the fix; the latest production workflow and live console were not rechecked for commit `48a6ae3`. EmailJS form delivery was not tested because that would send an external message.

What to remember

- Resolve imports from the importing module and CSS assets from the stylesheet.
- Expect localStorage to be empty on a different origin and in a fresh browser.
- Guard optional data before property access; keep unrelated UI behavior available.
- Use the Console and Network panel together when a page appears to load but controls do nothing.
- Do not claim a production fix until the deployed version is checked.

Review questions

1. Why did the old module path request `/Shared/…`?
Because `javascript2.js` is at the project root and `../../` climbs above the Pages repository path.

2. Why was the same localStorage record missing on GitHub Pages?
Because localhost and the GitHub Pages site have different origins, and the browser isolates storage per origin.

3. Why check `user` before assigning `user.AiVoice`?
The storage key may be absent, in which case parsing returns `null`; the guard prevents a property-access exception.

4. What proves the profile asset path is correct?
The filename and extension match the tracked asset, and the corrected local profile browser run had no failed asset request.

5. What is still needed to verify production?
Confirm the latest Pages workflow completed for commit `48a6ae3`, then reload the deployed site and inspect Console/Network while testing a task interaction.

> **KEY IDEA**
>
> Good debugging connects a symptom to evidence: a failed request, a specific code assumption, a small correction, and a repeatable verification.

Prepared from the committed diff and browser observations. No other files were changed to create this guide.